# FINTRUST DIGITAL BANK

## Week 4 Final Data Analytics Report

AnalystLab Africa Experience Lab | Test • Validate • Refine • Present

# Executive Summary

This report consolidates the Week 1–3 FinTrust Data Analytics work into a Week 4 final-review package. The visible sample contains 33 customer records and 33 transaction records with a total transaction value of NGN 959,180.43. The most important conclusion is a data-linkage limitation: only 1 of 33 transaction Customer_IDs match the visible customer export. Therefore, channel and transaction-type findings can be reported for the visible transaction sample, but customer-level transaction conclusions cannot be treated as reliable.

# 1. Scope and Method

The analysis continues the existing FinTrust project and uses the structured CSV transcriptions included in the Week 3 package. It applies reproducible Pandas checks, SQL review queries, aggregation reconciliation, duplicate and missing-value checks, customer-key matching, outlier analysis and documentation of field/time coverage. The source datasets are synthetic educational data. The transcribed CSVs are not original raw system extracts.

# 2. Final KPI Snapshot

| KPI | Validated result |
|---|---|
| Customer records | 33 |
| Transaction records | 33 |
| Total transaction value | NGN 959,180.43 |
| Mean transaction value | NGN 29,066.07 |
| Median transaction value | NGN 6,702.14 |
| Customer ID match rate | 3.0% (1/33) |
| IQR high-value rows | 5 above NGN 53,278.94 |

# 3. Final Business Insights

• Mobile App is the highest-value channel in the visible sample: NGN 493,341.73 (51.4% of transaction value). This supports monitoring app-channel performance, but not claims about satisfaction or growth drivers.

• Cash Withdrawal is the highest-value transaction type: NGN 395,660.05 (41.2%). This is descriptive of the sample only.

• Transaction values are right-skewed: the mean (NGN 29,066.07) exceeds the median (NGN 6,702.14); the median is a more representative central-value measure for this sample.

• 5 transactions exceed the IQR upper fence of NGN 53,278.94 and contribute 71.4% of sample value. They are statistical outliers, not automatic evidence of errors or fraud.

• Customer transaction linkage is unreliable: 1/33 transaction IDs match the customer export. Resolve this issue before reporting customer-level transaction KPIs.

# 4. Final Validation and Testing

The companion Excel workbook contains the detailed test matrix, findings, KPI definitions and recommendations. Recomputed checks confirm row counts, total value, means/medians, channel/type aggregation and IQR outlier calculations. The customer key join fails the reliability gate; the correct action is to document and block customer-level attribution rather than force a match.

# 5. Management Recommendations

• Prioritise channel monitoring — Mobile App contributes the largest value share in the sample. Recommended action: Review channel metrics regularly after confirming larger and more representative coverage. Caveat: Do not infer satisfaction or adoption drivers from this small export alone.

• Investigate high-value records through a controlled QA process — Five IQR outliers account for 71.4% of sample value. Recommended action: Validate source records, business context and applicable controls before operational decisions. Caveat: IQR outliers are not automatically suspicious or fraudulent.

• Resolve customer key mapping — Only 1/33 transaction customer IDs match. Recommended action: Obtain a corrected customer-key mapping and rerun customer-level analysis. Caveat: Do not publish customer-level transaction KPIs until this is resolved.

• Request transaction status and risk fields — The visible transaction export does not contain Transaction_Status or Risk_Review_Flag. Recommended action: Obtain approved fields before creating success/failure or risk-review visuals. Caveat: Do not fabricate these metrics or treat the synthetic target as real fraud detection.

• Obtain wider date coverage — Visible transaction dates range from 2026-01-01 00:00:00 to 2026-01-01 05:45:00. Recommended action: Use a representative multi-day/month dataset before drawing trend conclusions. Caveat: Hourly patterns in this export are exploratory only.

# 6. Limitations and Responsible Use

• The dataset is synthetic and intended for education, not production banking decisions.

• The visible transaction export does not include Transaction_Status or Risk_Review_Flag; success/failure and risk-review KPIs cannot be calculated from the visible fields.

• The visible transaction timestamps range from 2026-01-01 00:00:00 to 2026-01-01 05:45:00, a short period that is insufficient for reliable long-term trend claims.

• Only 1 of 33 transaction Customer_IDs matches the customer export; customer-level joins must be treated as unresolved.

• A native Power BI .pbix was not generated here. The package includes a build guide and validation targets; the actual dashboard must be built, saved, reopened and tested in Power BI Desktop.

• No claim is made that outputs from other internship tracks were integrated unless actual artifacts and evidence are available.

# 7. Week 4 Completion Status

| Deliverable | Status |
|---|---|
| Final SQL script | Prepared in package |
| Python notebook | Week 3 notebook included; rerun in the target environment as a final reproduci |
| KPI and validation workbook | Prepared |
| Final report and recommendations | Prepared |
| Power BI native .pbix | Outstanding — build and test in Power BI Desktop |
| Final video presentation | Presentation deck and speaking script prepared; user must record and submit |
| GitHub and social posts | User must publish/update and verify links |

Prepared as an educational portfolio artifact. Confirm all final outputs against the official Week 4 assignment and actual work completed before submission.